CIC-IDS2017 Zero-Day Anomaly Detection: Phase 1 ML Baselines and Phase 2 Deep Autoencoder

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Abstract

This report presents a two-phase anomaly-detection pipeline for CIC-IDS2017 with an explicit held-out-family setup to simulate a zero-day condition. Phase 1 uses classical unsupervised models (Isolation Forest and One-Class SVM), while Phase 2 uses a deep autoencoder with reconstruction-error thresholding. Across executed notebook outputs, the deep model provides stronger discrimination on known attacks than the ML baselines. At the same time, all models show near-perfect AUC on the held-out Heartbleed split but very low precision, which indicates threshold instability under severe class imbalance. We therefore treat the zero-day result as promising but not conclusive without stronger robustness validation.

1 Problem framing and motivation

Signature-based IDS methods are effective when attack patterns are already known, but they can miss novel or evolving traffic behaviors. For this reason, this work is framed as *benign-only anomaly detection*: models learn normal flow structure and flag deviations at inference time.

The practical objective is not just to maximize one metric, but to compare whether a deep reconstruction model provides a meaningful gain over lighter unsupervised baselines under the same split protocol, feature set, and thresholding rule. This framing is important because, in security deployment, a small gain in ranking quality can still be unacceptable if it produces unstable precision under realistic imbalance.

2 Dataset protocol and distribution observations

Both notebooks use the same CIC-IDS2017 8-day CSV bundle and keep numeric flow features only (78 dimensions). Labels are reduced to binary anomaly targets while preserving family labels for split control.

The executed logs show strong distribution asymmetry: benign traffic dominates training, while attack volume is concentrated in a few families (mainly DoS, PortScan, DDoS). The held-out Heartbleed family has only 11 rows, which means any threshold-sensitive metric can fluctuate sharply. This is exactly why very high zero-day AUC alone should not be over-interpreted.

From a modeling perspective, the data behavior suggests:

- multi-modal anomaly structure across families,
- high imbalance between normal and attack classes,
- potential train-test mismatch risk if benign traffic drifts over time.

3 Assumptions and methodological limits

The current pipeline assumes approximately i.i.d. sampling for train/validation partitions and quasi-stationary feature generation between training and inference. In operational IDS streams,

these assumptions are often violated due to seasonal traffic changes, endpoint churn, and changing attack tooling.

A second assumption is threshold transferability: the 99th-percentile threshold selected on benign validation is applied to mixed test streams. This is simple and reproducible, but it introduces bias-variance tension: tighter thresholds improve recall while sharply reducing precision under imbalance.

4 Methods

4.1 Phase 1: ML baselines

Phase 1 standardizes benign training features and fits two models:

- Isolation Forest (500 trees, contamination 0.01),
- One-Class SVM (RBF kernel, $\nu = 0.03$, $\gamma =$ `scale`).

For both, anomaly cutoffs are derived from the 99th percentile of benign validation scores. This creates a consistent thresholding policy across known-attack and held-out-family tests.

4.2 Phase 2: deep autoencoder

Phase 2 uses a feed-forward autoencoder with hidden layers (128, 64), bottleneck 32, MSE loss, and Adam optimization. The best validation checkpoint is selected at epoch 49 in the recorded run. Detection uses reconstruction error against the same percentile-based benign thresholding principle as Phase 1.

Compared to one-class boundary fitting, reconstruction modeling can capture higher-order feature interactions, which helps explain why Phase 2 improves known-attack AUC and F1 in this run.

5 Results from executed notebook outputs

Table 1: Phase 1 metrics

Model / Split	ROC-AUC	Precision	Recall	F1
Isolation Forest (known)	0.6984	0.9329	0.0144	0.0284
Isolation Forest (zero-day)	0.9996	0.0186	1.0000	0.0366
One-Class SVM (known)	0.7588	0.9972	0.4124	0.5835
One-Class SVM (zero-day)	0.9995	0.0168	1.0000	0.0330

Table 2: Phase 2 metrics

Model / Split	ROC-AUC	Precision	Recall	F1
Autoencoder (known)	0.8302	0.9946	0.4331	0.6035
Autoencoder (zero-day)	0.9994	0.0083	1.0000	0.0166

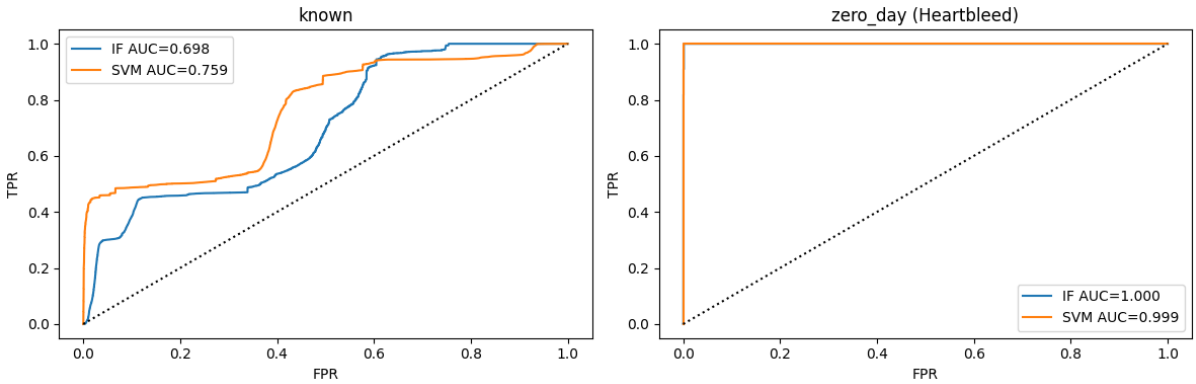

Figure 1: Phase 1 ROC curves from executed notebook output.

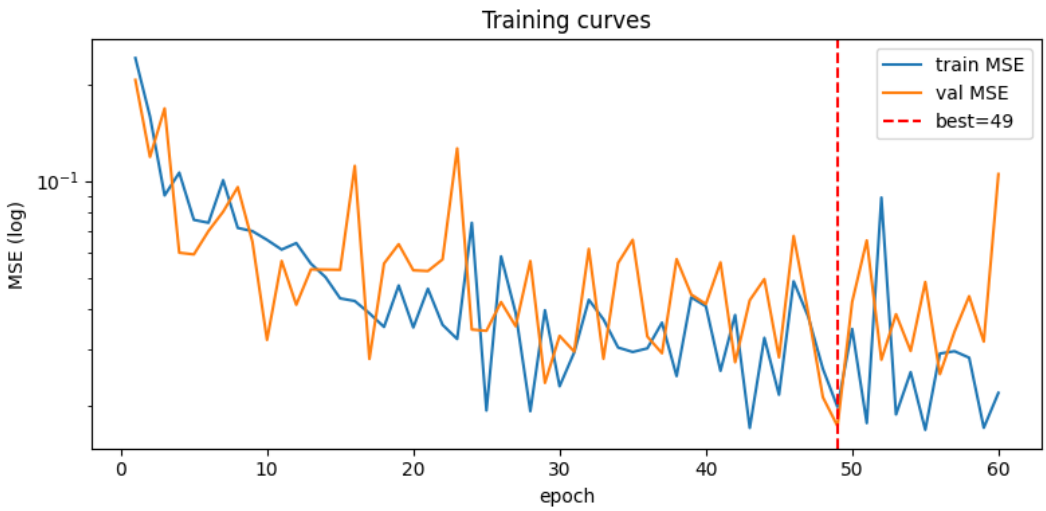

Figure 2: Phase 2 training and validation reconstruction-loss curves from executed notebook output.

6 Interpretation

The known-attack comparison is the clearest result: the autoencoder improves both ranking quality and balanced detection utility over the two classical baselines in this run. However, the held-out Heartbleed split demonstrates that very high AUC can coexist with extremely low precision if positive support is tiny and threshold transfer is rigid.

Therefore, this report treats the current zero-day evidence as *directionally positive* but not deployment-ready. To increase technical confidence, the next iteration should add day-wise out-of-distribution testing, threshold calibration analysis under shifted benign traffic, and family-level error slicing.

7 Conclusion

This two-phase CIC-IDS2017 implementation is functional, reproducible from notebook artifacts, and analytically stronger than a single-model baseline comparison. The deep phase provides better known-attack discrimination, while zero-day claims remain constrained by data sparsity and calibration sensitivity. The immediate next step is robustness-first validation, not additional architecture complexity.